

Run 09 | NME

Global (including India)

IV + controls

Target Necessity-motivated entrepreneurship; binary 1 vs. 0

Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022

Positive cases 69,931 (5.21%) **Case time** 43m 37s

Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)

Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.9793). Highest F1: Random Forest (0.6629). For Random Forest, adding controls changed ROC-AUC by +0.0168 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.9793	0.7143	96.23	62.07	71.14	0.6629	2m 39s
XGBoost	0.9791	0.7095	96.15	61.33	70.74	0.6569	23s
CatBoost	0.9788	0.7060	96.14	61.21	70.39	0.6548	2m 49s
H2O GBM	0.9772	0.6827	95.99	59.99	69.13	0.6423	48s
Elastic Net	0.9744	0.6484	95.33	53.89	71.64	0.6151	11m 58s
Lasso	0.9744	0.6485	95.33	53.88	71.61	0.6149	10m 36s
Logistic Regression	0.9742	0.6474	95.33	53.89	71.45	0.6144	1m 38s
Ridge	0.9731	0.6315	95.65	57.24	65.14	0.6093	8m 11s
Decision Tree	0.9684	0.5885	96.17	66.94	52.27	0.5871	11s
Naive Bayes	0.9242	0.3414	91.28	33.94	71.20	0.4597	3m 39s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	16,662.78	28.04
2	Established business owner	7,720.79	12.99
3	Survey year	4,847.78	8.16
4	Occupational status	4,057.45	6.83
5	GDP per capita (PPP)	3,684.25	6.20
6	Population	3,065.81	5.16
7	GDP growth	3,017.06	5.08
8	Age	2,882.01	4.85

Share is the percentage of total forest impurity importance across all 40 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Source: Completed Objective 1 batch, 27 September 2026; 09_obj1_global_iv_cv_NME. Scores and importance are taken directly from saved CSV outputs.